Sankar Dev Santhosh

Applied AI Engineer · NLP · RAG Systems · Transformer Models · MLOps

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Work Authorisation: Eligible to work in Germany

PROFESSIONAL SUMMARY

Applied AI Engineer with **3+** years of experience building production NLP systems, transformer pipelines, and retrieval-augmented AI applications across research and deployment environments. Published NLP researcher with transformer-based NLP research achieving **95.08% F1-score**, published in a **Q1-ranked Scopus-indexed journal** (JDSIS, 2026). Focused on privacy-preserving AI systems compliant with the EU AI Act and GDPR.

FEATURED PROJECT

KlarKI — Local-First EU AI Act & GDPR Compliance Auditor

2026

Python · FastAPI · gBERT · ONNX · NVIDIA Triton · ChromaDB · spaCy · LangGraph · Docker · React

- Architected a **7**-container local-first bilingual AI compliance platform for EU AI Act + GDPR auditing, designed for German SMEs with zero external API calls and full on-device inference.
- Built an end-to-end NLP pipeline covering document ingestion → actor classification → Annex III applicability gating → hybrid RAG retrieval → deterministic multi-agent LLM reasoning → scored PDF/JSON compliance reports.
- Fine-tuned **spaCy NER** and **deepset/gbert**-base classifiers, deployed via **ONNX + NVIDIA Triton**, achieving **50–100×** faster inference with reliable fallback through deterministic regex, NLI validation, and ML decision trees.
- Improved compliance evidence retrieval by replacing basic **RAG** with hybrid retrieval using BM25 + ChromaDB + Reciprocal Rank Fusion + cross-encoder re-ranking, significantly reducing hallucination risk and improving audit defensibility.
- Built **200+** automated tests covering unit logic, golden dataset validation, RAG retrieval quality, hallucination detection, and adversarial regulatory cases.

Results: Macro F1 ≥ 0.95 · NER F1 ≥ 0.95 · RAG Precision@3 ≥ 85%

PROFESSIONAL EXPERIENCE

Research Assistant — NLP & Multimodal AI

Sep 2024 – Present

University of East London · London, UK

- Built scalable NLP data pipeline processing **250,000+** text samples from GoEmotions and Hate Speech corpora.
- **Fine-tuned RoBERTa** (PyTorch, Hugging Face) for multi-class emotion recognition; achieved 95.08% F1-score, published in JDSIS (2026).
- Reduced model training latency by **~20%** through batching optimisations across classical and transformer-based models.
- Developed **SoundSense**, a multimodal vocal anomaly detection system using Wav2Vec2.0; achieved **78.10% accuracy** on CREMA-D, outperforming SVM/RF baselines by **30%.**

Software Engineer

Feb 2023 – Oct 2023

Techise Solutions · India

- Developed backend systems using **Laravel** and **MySQL** for multiple client applications across diverse industry sectors.
- Built responsive frontends using React, improving cross-device usability.

Software Associate

May 2022 – May 2023

Riss Technologies · India

- Contributed to delivery and maintenance of **200+** bespoke software solutions across Python, PHP, and React stacks.
- Deployed predictive ML models reducing manual data processing and inference optimisation time by **~20%.**

TECHNICAL SKILLS

ML & AI	PyTorch, TensorFlow, Hugging Face, BERT, RoBERTa, Wav2Vec2.0, LangChain, RAG, Multimodal AI, spaCy
MLOps & Deploy	Docker, FastAPI, NVIDIA Triton, ONNX Runtime, CI/CD, AWS EC2, Git, WebSockets, REST APIs
Data & Infra	ChromaDB, PySpark, MongoDB, Redis, SQL
Tech Stack	Python, TypeScript, JavaScript, React, Next.js, Node.js, Django, Flask
AI Governance	EU AI Act, GDPR / DSGVO, Privacy-Preserving Architectures

PUBLICATION

Journal of Data Science and Intelligent Systems (JDSIS), Q1-ranked (Scopus indexed, SJR)

2026

NLP Framework to Safeguard Youngsters Online Using Advanced Transformer-Based Models

EDUCATION

MSc Artificial Intelligence (with Industrial Placement) 2023 – 2025

University of East London · London, UK

Pass with **Distinction** · Equivalent to German grade 1.0 – 1.3 · Key modules: Artificial Intelligence 98/100 · Intelligent Systems 86/100 · Machine Learning on Big Data · Big Data Analytics

BSc Computer Applications 2019 – 2022

Kerala University · India

SELECTED PROJECTS

VoxChain — Real-Time LLM Voice Agent

LangChain · Qwen LLM · FastAPI · WebSockets

- Built a real-time voice AI system with bidirectional streaming (speech ↔ LLM ↔ audio).
- Deployed end-to-end agentic pipeline enabling low-latency conversational interaction.

SmartSafe — Multimodal Liveness Detection

Python · OpenCV · face_recognition · Audio Processing

- Designed biometric attendance system combining facial recognition with audio liveness detection.
- Generated multimodal confidence score to mitigate replay and spoofing attacks.

EmoTract — 28-Class Real-Time Emotion Classifier

BERT · RoBERTa · Random Forest · Django · React · Docker

- Built multi-label emotion classifier predicting 28 emotion categories from real-time text input.
- Deployed full-stack system from fine-tuned transformer model to containerised web interface.

ADDITIONAL INFORMATION

Languages	English (C1 – Professional) · German (A1 – A2, actively improving) · Malayalam (Native)
Relocation	Open to Germany — **available immediately**